1. What is the purpose of a loop structure?

The purpose of a loop structure is to repeat a series of instructions while a specific condition is true. Loops are useful for tasks like adding numbers, repeating calculations, or asking users to provide accurate data.

2. Explain the difference between a while statement and a do-while statement.

Since the while statement checks the condition before completing the function, it might not run at all if the condition is initially false. Before determining its condition, a do-while statement executes the code at least once.

3. An input validation loop is a loop that checks user input for valid data. If valid data is not entered, the loop iterates until valid data is entered. In which review of this chapter did you write code for an input validation loop?

Until the game reaches a stopping condition, the software continues requests input from the user using the input validation loop found in the WordGuess. It also uses a do-while loop that requests that the user input a number smaller than 4 in order to illustrate input validation.

4. a) What is an infinite loop?

An infinite loop is a loop that continues running forever because its stopping condition never becomes false.

b) List two types of errors that can lead to an infinite loop.

Two types of errors that can cause an infinite loop are syntax errors and logic errors.

c) What is meant by overflow?

Overflow happens when a number gets too large or too small to be stored in the available number of bits. This can cause the stored value to change unexpectedly.

5. How many times will the do-while loop execute?

```
int x = 0;
do {
   x = x + 2;
} while (x < 120);
```

The loop will execute 60 times. The value of x increases by 2 each time, starting at 0, until it reaches 120. Then, the condition becomes false and the loop stops.

6. What initial value of x would make the loop infinite?

```
do {
   x = x + 3;
} while (x < 120);
```

As mentioned, because x has no initial integer value, this Java loop is indefinite. When the loop increases x by 3 and ends up reaching or passing 120, the condition is false. Even if there is an integer overflow, the value will eventually reach the halting condition.

7. Compare and contrast counters and accumulators. List two uses for each.

A counter keeps track of how many times something happens and usually increases by 1 each time. An accumulator adds values together and can increase by different amounts.

Two uses for counters are:

- Keeping track of the number of guesses in a game.
- Counting how many values a user enters.

Two uses for accumulators are:

- Calculating the total of a group of numbers.
- Keeping a running total of points earned in a game.

8. Write a for statement that sums the integers from 3 to 10, inclusive.

```
int sum = 0;

for (int i = 3; i <= 10; i++) {
   sum += i;
}

System.out.println(sum);
```

The output is 52 because the program adds all the integers from 3 through 10.

9. List two factors that should be considered when determining which loop structure to choose.

Two factors to consider are:

- Whether the loop needs to execute at least once.
- Whether the number of repetitions is known in advance or depends on a condition being met.

10. a) List two methods for debugging an application.

Two methods for debugging an application are:

- Using additional println() statements to display variable values and check what the program is doing.
- Stepping through the code using a debugger to examine variables and find where an error occurs.

b) Which method listed in part (a) would be best for debugging an application that generates random numbers that are used for determining the condition of an if statement? Why?

It is better to use more println() statements since they can show which condition the program is checking as well as the random number generated. This makes it simpler to understand why the if statement takes a certain course, especially when the random number varies with each program execution.